Early Multimodal Physiological Phenotypes and Outcomes in Critically Ill Adults With Non-Traumatic Subarachnoid Hemorrhage: A Retrospective Cohort Study Using MIMIC-IV 3.1

Abstract

Background

Non-traumatic subarachnoid hemorrhage (NSAH) is a high-acuity neurocritical condition characterized by significant clinical heterogeneity and variable risk of systemic complications. Traditional severity assessment tools focus predominantly on neurological impairment, overlooking systemic physiological derangements. We aimed to identify early, clinically interpretable, multimodal physiological phenotypes in critically ill NSAH patients using unsupervised machine learning, and to characterize their association with in-hospital outcomes and early anemia.

Methods

We conducted a retrospective cohort study using the MIMIC-IV 3.1 database, with external validation in the eICU Collaborative Research Database (eICU-CRD). Adult patients admitted to the intensive care unit (ICU) with NSAH and an ICU stay of at least 24 hours were eligible. Using eight core physiological variables (neurological, circulatory, respiratory, renal, hematologic, and coagulation markers) captured within the first 48 hours of ICU admission, we applied a clustering pipeline consisting of log1p-transformation (for skewed features), Z-score standardization, Principal Component Analysis (PCA), and K-means clustering. The primary outcome was in-hospital mortality. We assessed phenotype associations with outcomes using multivariable logistic regression and Cox proportional hazards models. External validation utilized a "frozen transport" approach to project external patients into the MIMIC-derived phenotypes.

Results

The development cohort included 1,186 patients (in-hospital mortality 19.81%; early anemia rate 26.56%; red blood cell [RBC] transfusion rate 0–48 h 2.02%). Three clinically interpretable phenotypes were identified: Phenotype 1 (P1: n=694, 58.5%; mild neurologic and systemic impairment), Phenotype 2 (P2: n=384, 32.4%; severe neurologic but mild systemic impairment), and Phenotype 3 (P3: n=108, 9.1%; severe neurologic and severe multisystem shock/organ dysfunction). In-hospital mortality rose monotonically: P1 (6.34%), P2 (32.55%), and P3 (61.11%). Compared with P1, P2 (adjusted odds ratio [aOR] 7.59, 95% CI 5.07–11.36, $p < 0.001$) and P3 (aOR 21.21, 95% CI 12.08–37.26, $p < 0.001$) remained associated with higher mortality in clinical-adjusted models. Anemia was enriched in P2 (41.41%) and P3 (66.67%) but was not independently associated with mortality (aOR 0.99, 95% CI 0.68–1.44, $p = 0.955$). In the external validation cohort (N = 843), the frozen MIMIC classifier preserved the mortality gradient (P1 5.4%, P2 25.7%, P3 42.7%) and aligned with independent APACHE scores (rho = 0.480, $p < 0.001$). Hb-free sensitivity validation models yielded similar conclusions for early anemia after phenotype adjustment ($p = 0.105$).

Conclusions

Unsupervised clustering of routine early ICU physiological data identifies three clinically distinct, transportable phenotypes in critically ill NSAH patients with progressive risk of mortality. These phenotypes capture systemic multi-organ involvement beyond neurological status alone, providing a framework for refined risk stratification and clinical trial design.

Introduction

Non-traumatic subarachnoid hemorrhage (NSAH) represents a critical and life-threatening neurological emergency, accounting for approximately 5% of all stroke cases, but carrying disproportionate morbidity and mortality, particularly in younger populations. Historically, clinical research and management strategies for NSAH have focused primarily on the primary intracranial insult, such as aneurysmal rupture, and subsequent neurological complications, including vasospasm, delayed cerebral ischemia (DCI), and hydrocephalus. Consequently, traditional risk-stratification tools, such as the Hunt and Hess scale and the World Federation of Neurosurgical Societies (WFNS) grading system, rely almost exclusively on neurological symptoms and the Glasgow Coma Scale (GCS).

However, critically ill adults with NSAH frequently suffer from severe extracerebral organ dysfunction. A massive sympathetic surge and hypothalamic-pituitary-adrenal axis activation following the initial hemorrhage trigger systemic inflammatory response syndrome (SIRS), neurogenic stunned myocardium, acute lung injury, renal hypoperfusion, anemia, and coagulation abnormalities. These systemic derangements are not merely secondary bystander events; they interact dynamically with the primary brain injury (organ cross-talk) and significantly influence both survival and functional recovery. Despite their clinical relevance, these systemic physiological dimensions are often evaluated in isolation, and their combined multimodal patterns remain poorly characterized.

Unsupervised machine learning algorithms, such as K-means clustering paired with dimensionality reduction, offer a powerful approach to uncover hidden structures in complex clinical data. By aggregating multiple physiological domains simultaneously, these methods can identify distinct patient-level subgroups, or "phenotypes," that reflect unique physiological signatures rather than isolated organ failure scores. Such phenotyping has successfully redefined risk stratification in general sepsis and acute respiratory distress syndrome, but its application to neurocritical care, specifically NSAH, remains limited. Furthermore, the role of early anemia—a common finding in NSAH that is frequently associated with poor outcomes—remains controversial, as it is unclear whether anemia represents an independent therapeutic target or a downstream marker of global physiological derangement.

To address these gaps, we aimed to identify early, clinically interpretable, multimodal physiological phenotypes in critically ill adults with NSAH and to evaluate their associations with outcomes and clinically relevant features. Using the Medical Information Mart for Intensive Care IV (MIMIC-IV 3.1) database as our development cohort and the eICU Collaborative Research Database (eICU-CRD) for external validation, we developed and validated a robust, transportable phenotyping pipeline. We hypothesized that NSAH patients could be classified into distinct multimodal physiological phenotypes with distinct survival trajectories and that accounting for these phenotypes would clarify the independent prognostic value of early anemia.

Methods

Data Sources

This retrospective cohort study utilized two large, multicenter, publicly available intensive care databases: the MIMIC-IV database (version 3.1) and the eICU Collaborative Research Database (eICU-CRD). MIMIC-IV contains high-resolution clinical data from over 190,000 intensive care unit (ICU) admissions at the Beth Israel Deaconess Medical Center (Boston, MA) between 2008 and 2022. The eICU database includes de-identified clinical data from over 200,000 ICU admissions across 208 hospitals in the United States between 2014 and 2015. Access to both databases was granted to the authors after completion of the Collaborative Institutional Training Initiative (CITI) training (Record ID: 57407421). The use of the databases was approved by the Institutional Review Boards of the Massachusetts Institute of Technology and the respective participating institutions. The studies are reported in compliance with the STROBE (Strengthening the Reporting of Observational Studies in Epidemiology) guidelines.

Cohort Selection

We constructed the primary development cohort from MIMIC-IV 3.1. Patients were eligible if they met the following inclusion criteria:

1. Admitted to the ICU with a primary or secondary diagnosis of non-traumatic subarachnoid hemorrhage, identified via International Classification of Diseases, Ninth Revision (ICD-9) code `430` or Tenth Revision (ICD-10) codes `I60.x`.
2. Adult patients (age ≥ 18 years at the time of admission).
3. ICU length of stay (LOS) of at least 24 hours to ensure adequate time for early physiological characterization.
4. Had complete or near-complete physiological data, defined as having no more than two missing values among the eight core physiological variables.

We excluded patients who:

1. Had explicit evidence of traumatic subarachnoid hemorrhage, identified by ICD-10 codes `S06.6x` or diagnostic text indicating trauma.
2. Had multiple ICU stays during their index hospitalization (only the first ICU stay was analyzed to prevent post-treatment physiological confounding).

3. Received massive red blood cell (RBC) transfusions within the first 24 hours of ICU admission (defined as ≥ 5 units of packed red blood cells), as this early, intensive intervention fundamentally alters baseline physiological signals.

For external validation, we applied identical inclusion and exclusion criteria to the eICU-CRD database, defining NSAH admissions via ICD-9 code `430`, apacheadmissiondx containing "subarachnoid hemorrhage", or specific diagnosis text entries, while excluding traumatic cases.

Feature Selection and Preprocessing

Eight core physiological variables were selected *a priori* to represent major homeostatic systems and organ functions relevant to NSAH prognosis:
1. **Neurological**: Minimum GCS motor score (`gcs_motor_min_48h`) within the first 48 hours. The GCS motor subscore was selected over the total GCS score to minimize the confounding effect of endotracheal intubation on verbal and eye components.
2. **Circulatory**: Minimum mean arterial pressure (`map_min_48h`) and maximum shock index (`shock_index_max_48h` , calculated as heart rate divided by systolic blood pressure) within the first 48 hours.
3. **Oxygenation**: Minimum pulse oximetry oxygen saturation (`spo2_min_48h`) within the first 48 hours.
4. **Renal**: Maximum creatinine level (`creatinine_max_48h`) within the first 48 hours.
5. **Coagulation**: Maximum International Normalized Ratio (`inr_max_48h`) within the first 48 hours.
6. **Hematologic**: Minimum hemoglobin level (`hb_min_48h_all`) and minimum platelet count (`platelet_min_48h`) within the first 48 hours.

The baseline/feature window was defined as the first 48 hours following ICU admission. For patients discharged or deceased between 24 and 48 hours, all available data up to the point of discharge or death were utilized. Clinical ranges were applied to all variables to exclude physiologically implausible outliers.

Missing values in the development cohort were extremely low: INR max had a missing rate of 5.48% (65/1,186), creatinine max had a missing rate of 0.08% (1/1,186), and all other core features had a 0.0% missing rate. Median imputation based on the development cohort was used to address missing values. Highly skewed variables, specifically creatinine and INR, were log1p-transformed (`log(x + 1)`) to achieve approximately symmetric distributions. All eight variables were then standardized to Z-scores (mean = 0, standard deviation = 1) using the StandardScaler logic.

Dimensionality Reduction and Clustering

To mitigate collinearity and noise among physiological features, we performed Principal Component Analysis (PCA) on the standardized eight-dimensional feature space. The number of principal components (PCs) retained was set to three based on eigenvalues and scree plot inspection. These 3 PCs accounted for 56.41% of the total explained variance (PC1: 30.27%, PC2: 13.77%, PC3: 12.36%).

Unsupervised clustering was performed in the 3-PC space using the K-means algorithm with a fixed random seed to guarantee reproducibility. The optimal number of clusters (K) was determined by evaluating the silhouette width, Calinski-Harabasz index, Davies-Bouldin index, minimum cluster size, and clinical interpretability. A K=3 solution was selected as the primary phenotyping model (the evaluation of alternative K selections is provided in Supplementary Figure 1). The resulting phenotypes were ordered P1, P2, and P3 by ascending overall severity and in-hospital mortality.

Statistical Analysis

Continuous baseline characteristics were described using medians with interquartile ranges (IQRs) and compared using the Kruskal-Wallis test. Categorical variables were expressed as frequencies (percentages) and compared using the Chi-square test or Fisher's exact test.

We constructed multivariable logistic regression models to assess the independent association between the identified phenotypes and the primary outcome of in-hospital mortality.

- **Model 1 (Clinical Main Effect)** adjusted for baseline demographics (age, sex), admission type, NSAH evidence level (based on ICD coding specificity), and the presence of an aneurysm diagnosis.
- **Model 2 (Process-of-Care Adjusted)** additionally adjusted for downstream ICU interventions and process-of-care markers (nimodipine administration, vasopressor use, mechanical ventilation, RBC transfusion, continuous renal replacement therapy [CRRT], external ventricular drain [EVD] or intracranial pressure [ICP] monitoring, and fluid balance) to examine the robustness of the phenotype-outcome association.

Time-to-event survival analysis was conducted using Kaplan-Meier curves, log-rank tests, and multivariable Cox proportional hazards models. Patients discharged alive were censored at their hospital discharge time.

Sensitivity Analyses

To evaluate the robustness of the primary clustering partition, we performed several sensitivity analyses:

1. **Complete-Case Analysis**: Re-running the pipeline after excluding all patients with any missing core physiological feature (N = 1,120).
2. **Strict Aneurysm Subgroup**: Restricting the analysis to patients with documented aneurysm diagnosis or securing procedures (clipping/coiling) (N = 763).
3. **ICU LOS ≥ 48 Hours**: Restricting the cohort to patients with ICU stay ≥ 48 hours to minimize survival/discharge bias (N = 1,005).
4. **Alternative Feature Windows**: Utilizing a 0–24-hour physiological window instead of 0–48 hours to reduce potential treatment contamination.
5. **Hemoglobin-Free and INR-Free Clustering**: Independently re-clustering after dropping hemoglobin or INR from the inputs to assess circularity in anemia adjustments and the impact of the most missing validation variable.
6. **High-Resolution Subtyping**: Exploring a K=4 partition to examine whether P3 could be refined further.
7. **Bootstrap Stability**: Re-sampling the development cohort 200 times to assess cluster assignment stability using the Adjusted Rand Index (ARI) (detailed in Supplementary Table 3 and visualized in Supplementary Figure 2).

External Validation (Frozen Transport)

External validation was performed in the eICU cohort (N = 843). Rather than refitting the clustering model, we applied a **Frozen Transport** approach:

1. eICU features were imputed using the frozen MIMIC-IV median values.
2. eICU creatinine and INR were log1p-transformed.
3. The data were standardized using the frozen MIMIC-IV StandardScaler mean and SD.
4. Standardized features were projected into the 3-PC space using the frozen MIMIC-IV PCA eigenvectors.
5. Patients were assigned to the closest MIMIC-derived phenotype centroid based on Euclidean distance.

We assessed the external validity of the transported phenotypes by evaluating: (a) the in-hospital mortality gradient across transported groups, (b) external criterion validation, correlating the phenotype assignments with independent eICU severity scoring systems (APACHE score, Acute Physiology Score [APS], and predicted mortalities) using Spearman's rank correlation, and (c) structural sensitivity, comparing the transported labels with an independent, de novo K-means (K=3) partition fit directly on the eICU data.

Results

Cohort Selection and Missingness

In MIMIC-IV, a total of 1,186 adult NSAH patients met all eligibility criteria and were included in the primary analysis (the baseline characteristics are presented in Table 1). The flowchart of patient selection is illustrated in Figure 1.

Figure 1. Cohort selection and analysis framework

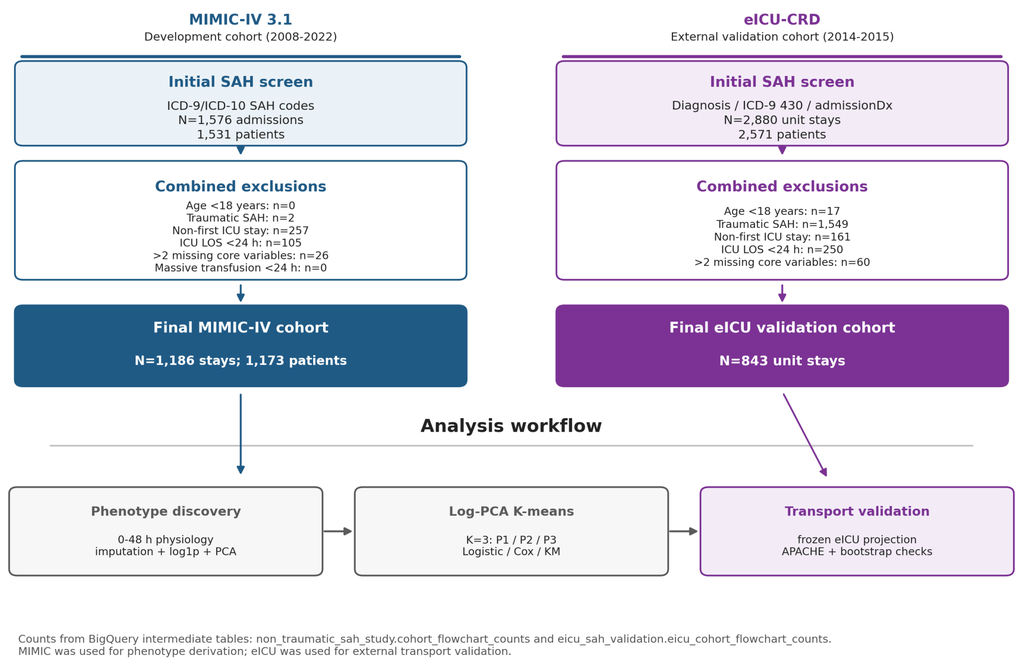

Figure 1. Cohort selection and analysis framework for the development (MIMIC-IV) and external validation (eICU) cohorts. Counts were generated from BigQuery intermediate cohort-flow tables; the lower panel summarizes the frozen MIMIC-derived analysis workflow used for phenotype discovery and eICU transport validation.

Core physiological data missingness was extremely low in the development cohort: INR max was missing in 5.48% (65/1,186) of patients, creatinine max in 0.08% (1/1,186), and all other features (hemoglobin, GCS motor, MAP, shock index, SpO2, platelets) had 0.0% missingness (feature missingness rates in both cohorts are detailed in Supplementary Table 1).

Primary PCA and Phenotype Profiles

PCA identified three principal components explaining 30.27%, 13.77%, and 12.36% of the variance, respectively (cumulative variance: 56.41%). The loadings of the eight physiological variables on the three PCs are detailed in Supplementary Table 2 and visualized in Supplementary Figure 5.

The primary K=3 solution partition separated the cohort into three distinct physiological phenotypes (Figure 2 and Table 2):

1. **Phenotype 1 (P1: n = 694, 58.5%)**: Characterized by mild neurologic and systemic physiological impairment. Patients had a median GCS motor score of 6.0 [IQR 5.0–6.0], median MAP of 64.0 mmHg [IQR 58.0–70.0], median hemoglobin of 11.70 g/dL [IQR 10.80–12.80], and normal renal and coagulation markers.
2. **Phenotype 2 (P2: n = 384, 32.4%)**: Characterized by severe neurological impairment but relatively mild systemic organ dysfunction. The median GCS motor score was 1.0 [IQR 1.0–4.0], but circulatory parameters, renal function (creatinine median 0.9 mg/dL), and coagulation (INR median 1.2) were only mildly deranged. Early anemia was moderately enriched (41.41%).
3. **Phenotype 3 (P3: n = 108, 9.1%)**: Characterized by severe neurological impairment combined with severe multisystem shock, hypoxemia, renal failure, and coagulopathy. The median GCS motor score was 1.0 [IQR 1.0–5.0], the median MAP was 55.0 mmHg [IQR 47.5–61.0], the median shock index was 1.16 [IQR 0.93–1.42], the median SpO2 was 88.5% [IQR 78.0–92.0], the median creatinine was 1.9 mg/dL [IQR 1.18–3.20], the median INR was 1.8 [IQR 1.40–2.30], and the median platelet count was 92.0 $\times 10^3/\mu L$ [IQR 40.75–160.25]. Early anemia was highly prevalent (66.67%).

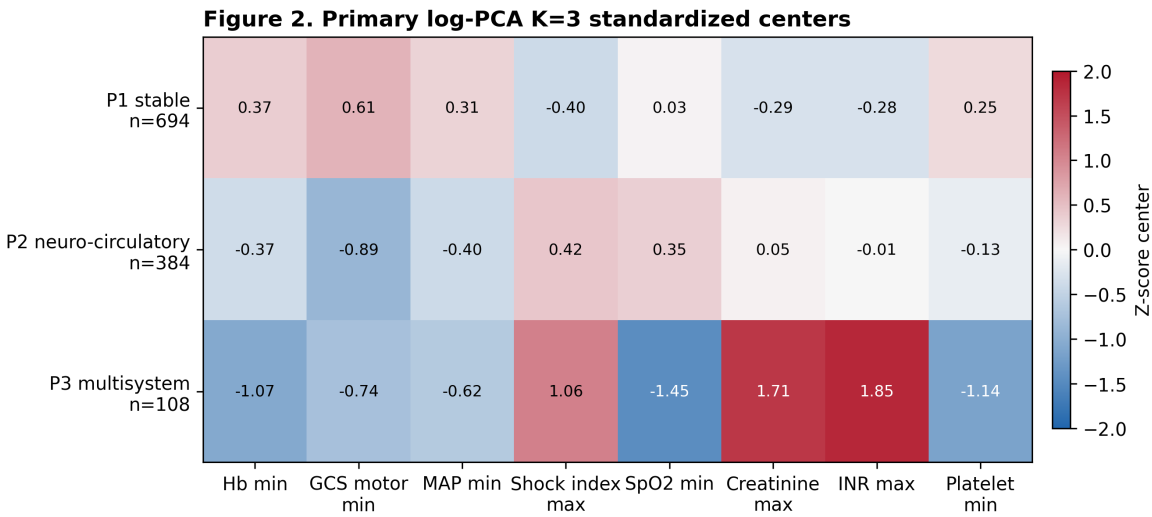

Figure 2. Early physiological profiles of the three identified phenotypes in MIMIC-IV. Values represent Z-score standardized cluster centers (Z-centers) for the eight core variables, with corresponding raw medians and IQRs.

Outcome Gradients and Survival Analysis

A strong, progressive outcome gradient was observed across the phenotypes in MIMIC-IV. In-hospital mortality rates were 6.34% (44 deaths) in P1, 32.55% (125 deaths) in P2, and 61.11% (66 deaths) in P3 (overall mortality rate 19.81%, 235 deaths; $p < 0.001$). ICU mortality showed a similar pattern: P1 (3.60%), P2 (26.56%), and P3 (50.93%). Median ICU length of stay was 6.38 days for P1, 10.56 days for P2, and 5.38 days for P3 (the shorter duration in P3 reflects early mortality, with 50.93% of patients dying in the ICU).

Unadjusted Kaplan-Meier survival curves revealed a significant survival separation among the three phenotypes (log-rank test, $p < 0.001$, Figure 3).

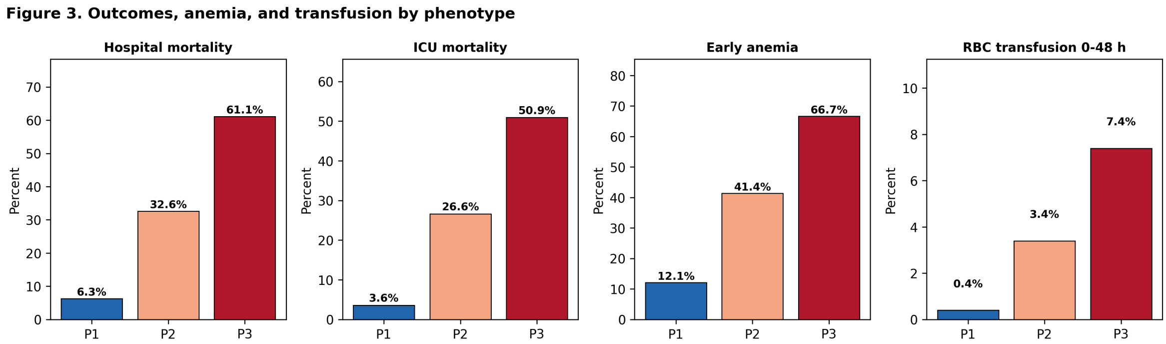

Figure 3. Outcome, anemia, and early RBC transfusion gradients across the three physiological phenotypes in the development (MIMIC-IV) cohort. Mortality and anemia are clinical outcomes/features; RBC transfusion is shown as a descriptive process-of-care marker and should not be interpreted as a treatment-effect estimate.

In unadjusted Cox proportional hazards models, both P2 (HR 4.20, 95% CI 2.97–5.94, $p < 0.001$) and P3 (HR 7.94, 95% CI 5.38–11.70, $p < 0.001$) had higher hazards of in-hospital death compared with P1.

Multivariable Regression and Process-of-Care Models

In the primary clinical adjusted logistic regression model (Model 1), the phenotypes remained independently associated with in-hospital mortality after adjusting for age, sex, admission type, NSAH evidence level, and aneurysm diagnosis (Table 3). Compared to P1, the odds of death were 7.59-fold higher for P2 (aOR 7.59, 95% CI 5.07–11.36, $p < 0.001$) and 21.21-fold higher for P3 (aOR 21.21, 95% CI 12.08–37.26, $p < 0.001$). Notably, early anemia was not independently associated with mortality (aOR 0.99, 95% CI 0.68–1.44, $p = 0.955$).

In the process-of-care adjusted model (Model 2), which included downstream ICU interventions as contextual severity and treatment-selection markers, the associations of P2 and P3 with in-hospital mortality were attenuated but persisted: P2 vs. P1 (OR 4.02, 95% CI 2.60–6.24, $p < 0.001$) and P3 vs. P1 (OR 11.75, 95% CI 6.52–21.20, $p < 0.001$). Nimodipine administration, vasopressor use, mechanical ventilation, RBC transfusion, CRRT, EVD/ICP monitoring, and fluid balance are reported as exploratory associations rather than treatment-effect estimates.

The adjusted Cox proportional hazards models showed a similar phenotype gradient (Table 4). In the process-adjusted exploratory survival model, P2 (adjusted HR 3.08, 95% CI 2.09–4.54, $p < 0.001$) and P3 (adjusted HR 5.24, 95% CI 3.25–8.47, $p < 0.001$) remained associated with higher in-hospital mortality. Because several process variables occurred during the 0–48 hour feature window, this model should be interpreted as a sensitivity analysis rather than a causal treatment model.

External Validation in eICU

The validation cohort from the eICU database comprised 843 eligible NSAH patients. Applying the frozen MIMIC-IV preprocessing parameters and cluster centroids projected the validation patients into the three phenotypes: P1 (n = 539, 63.9%), P2 (n = 222, 26.3%), and P3 (n = 82, 9.7%).

The external validation cohort demonstrated a monotonic outcome gradient across the transported phenotypes (Table 5):
- **Hospital Mortality**: P1 (5.4%), P2 (25.7%), P3 (42.7%)
- **ICU Mortality**: P1 (2.8%), P2 (16.2%), P3 (29.3%)
- **Early Anemia**: P1 (11.4%), P2 (34.5%), P3 (58.0%)
- **RBC Transfusion 0–48 h**: P1 (0.4%), P2 (5.9%), P3 (11.0%)

External criterion validation against independent eICU severity measures demonstrated concordant severity gradients (Figure 4). These scores were not used for clustering or transport assignment. The transported phenotype order aligned with:
- **APACHE Score**: P1/P2/P3 medians of 36 / 57 / 79 (Spearman rho = 0.480, $p < 0.001$)
- **Acute Physiology Score (APS)**: P1/P2/P3 medians of 27 / 49 / 67 (Spearman rho = 0.508, $p < 0.001$)
- **Predicted Hospital Mortality**: P1/P2/P3 medians of 0.069 / 0.243 / 0.426 (Spearman rho = 0.445, $p < 0.001$)

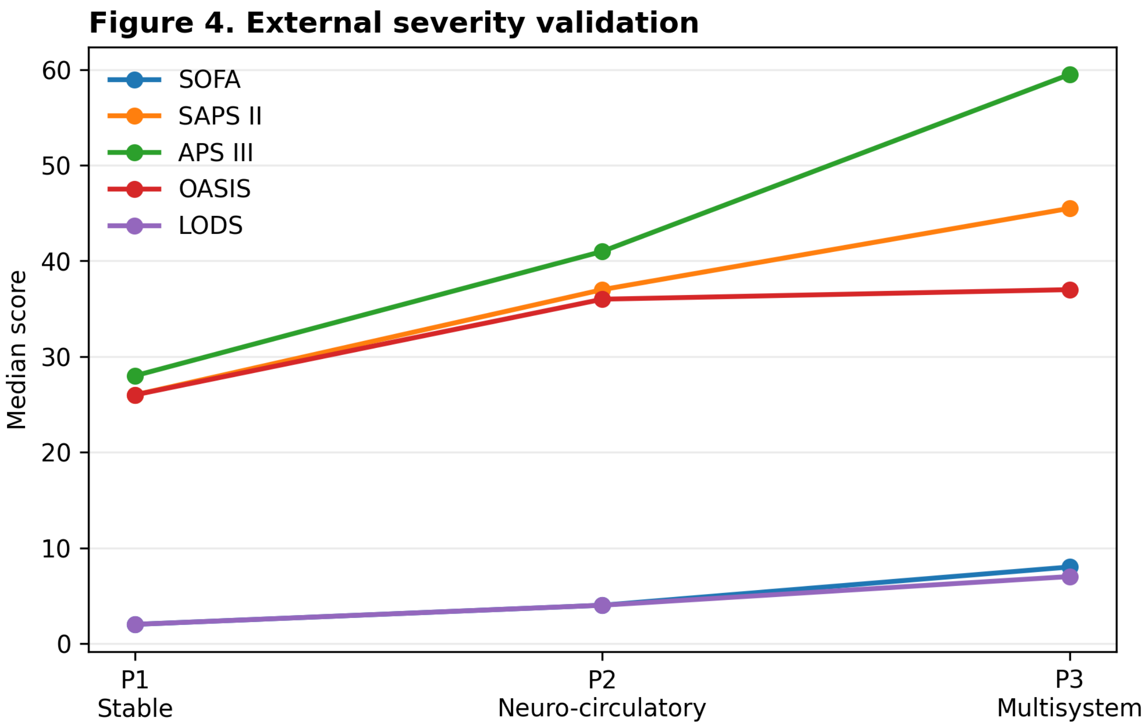

Figure 4. External criterion validation of transported phenotypes in the eICU cohort (N=843). APACHE/APS scores and predicted mortalities were not clustering inputs; they were used only to assess whether the transported phenotypes aligned with independent severity measures.

To address the potential circularity of adjusting for anemia using a phenotype that contains hemoglobin, we performed multivariable logistic regression in eICU using the **Hb-free transport model**. After adjusting for age, gender, and the Hb-free phenotype, the transported phenotypes remained associated with mortality, whereas early anemia remained non-significant:

- P2 vs. P1: OR = 6.16 (95% CI 3.74–10.13, $p < 0.001$)
- P3 vs. P1: OR = 10.27 (95% CI 5.68–18.57, $p < 0.001$)
- Early anemia: OR = 1.47 (95% CI 0.92–2.36, $p = 0.105$)

Independent *de novo* K-means (K=3) clustering in the eICU cohort recovered a similar risk gradient (hospital mortality: cluster 1 = 5.8%, cluster 2 = 18.1%, cluster 3 = 42.0%). However, patient-level concordance between transported and *de novo* labels was extremely low (Adjusted Rand Index [ARI] = -0.003, Normalized Mutual Information [NMI] = 0.002, same ordered label rate = 43.9%). This structural sensitivity analysis supports the transportability of the risk gradient, while also showing that exact patient-level partition boundaries are sensitive to database-specific feature distributions (detailed calibration and comparison metrics are in Supplementary Figure 7).

Model Incremental Prediction and SHAP Importance

Logistic regression models showed that incorporating the phenotypes improved prediction of in-hospital mortality compared with a GCS-only model (Figure 5). The 8-variable model achieved an AUROC of 0.842 in cross-validation (Brier score: 0.119), compared with 0.539 for the GCS-only model (Brier score: 0.150). The phenotype-only model achieved an AUROC of 0.754.

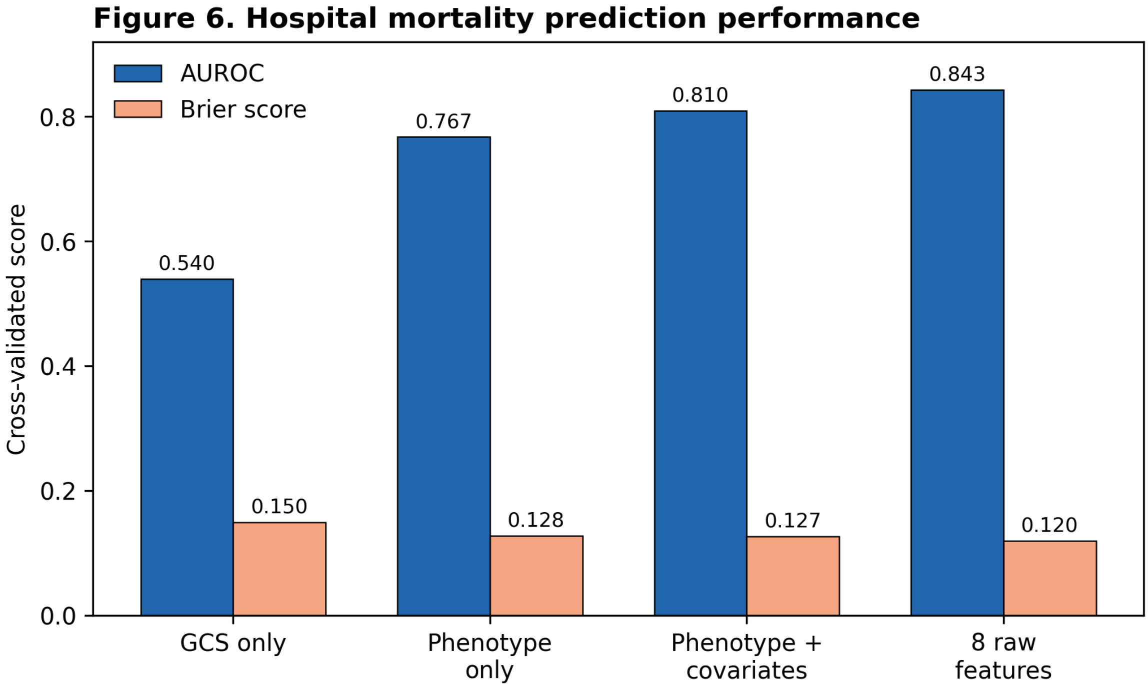

Figure 5. Incremental prediction of in-hospital mortality in MIMIC-IV, comparing GCS-only, phenotype-only, and multivariable physiological models.

SHAP-style attribution analysis from the standardized mortality logistic model identified the minimum GCS motor score as the most influential predictor (coefficient = -0.97, SHAP rank 1), followed by maximum creatinine (coefficient = 0.38, rank 2) and minimum platelets (coefficient = -0.21, rank 3) (see Supplementary Figure 6 for the odds and hazard ratios across models).

Sensitivity and Robustness Analyses

The phenotype risk gradient was remarkably stable across all pre-specified sensitivity analyses (Table 6).

- In the **complete-case subcohort** (N = 1,120), mortality rates remained P1 (6.53%), P2 (38.58%), and P3 (65.07%).

- In the **strict aneurysm subgroup** (N = 763), mortality rates were P1 (5.73%), P2 (30.82%), and P3 (50.00%).
- Excluding patients with early RBC transfusions or restricting to ICU stays ≥ 48 hours did not alter the monotonic risk separation.
- drops in hemoglobin and INR (INR-free and Hb-free clustering) preserved the mortality gradients, with ARI concordance rates of 0.631 and 0.736 against the primary model (with cluster profiles illustrated in Supplementary Figure 3).
- High-resolution K=4 clustering partitioned P3 into two smaller, extremely high-risk subgroups (P3: n = 39, mortality 61.54%; P4: n = 39, mortality 56.41%), showing that the K=3 severe phenotype represents the unified tail of the risk distribution (Supplementary Figure 4).
- Bootstrap validation over 200 iterations yielded a mean ARI of 0.920 (std = 0.048, min = 0.745), indicating high mathematical stability (Supplementary Table 3 and Supplementary Figure 2).

Discussion

Principal Findings

We successfully identified and externally validated three early, clinically distinct physiological phenotypes (P1, P2, P3) in critically ill adults with non-traumatic subarachnoid hemorrhage (NSAH). The primary findings are summarized as follows:

1. Routine physiological data collected during the first 48 hours of ICU admission can reliably partition NSAH patients into three phenotypes characterized by distinct patterns of neurological and systemic organ dysfunction.
2. These phenotypes display a robust, monotonic risk gradient for in-hospital and ICU mortality, as well as ICU and hospital length of stay.
3. The phenotype-outcome association persisted after adjusting for demographic factors, hemorrhage etiology, and intensive process-of-care markers.
4. While early anemia is enriched in the more severe phenotypes, it is not an independent predictor of in-hospital mortality once the global physiological phenotype is accounted for.
5. The phenotype classifier transported to the eICU database with preserved mortality gradients and concordance with independent APACHE severity scores.

Clinical and Physiological Interpretation

The three identified phenotypes provide a comprehensive representation of NSAH patients in the real-world ICU setting. **Phenotype 1 (P1, 58.5% of cohort)** represents the "mild neurologic/systemic" group. These patients are neurologically intact (GCS motor median of 6) and have stable cardiopulmonary, renal, and coagulation profiles. They have a very low risk of mortality (6.34%) and likely represent patients with low-grade Hunt-Hess/WFNS scores, or those with perimesencephalic non-aneurysmal hemorrhages who require minimal organ support.

Phenotype 2 (P2, 32.4% of cohort) represents the "severe neurologic/mild systemic" group. These patients present with profound neurological depression (median GCS motor of 1.0) but lack severe circulatory failure, renal impairment, or severe coagulopathy. Their mortality rate is substantial (32.55%), driven primarily by the primary intracranial injury.

Phenotype 3 (P3, 9.1% of cohort) represents the "severe neurologic/severe multisystem" group. Patients in this category have severe neurological injury (median GCS motor of 1.0) coupled with systemic cardiovascular shock (median MAP of 55 mmHg, shock index of 1.16), hypoxemia (median SpO2 of 88.5%), renal failure (median creatinine of 1.9 mg/dL), and coagulopathy (median INR of 1.8, platelets of 92.0 $\times 10^3/\mu L$). This group represents a highly vulnerable subcohort experiencing extensive systemic organ activation, sympathetic storm, and multi-organ dysfunction syndrome (MODS). P3 carries an extremely high mortality rate (61.11%).

Understanding these phenotypes goes beyond the traditional Hunt-Hess or WFNS classification. For example, while both P2 and P3 share a median GCS motor score of 1.0, they represent entirely different physiological profiles and mortality risks (32.55% vs 61.11%). A patient in P2 suffers from localized neurological failure, whereas a patient in P3 suffers from systemic shock and multi-organ exhaustion. This distinction is critical because global scores like the GCS cannot differentiate between these two states, yet their resource needs, complication profiles, and prognoses are vastly different.

Relationship with Existing Severity Scores and Literature

Traditional clinical assessments in NSAH are dominated by neurological scales, which are known to suffer from inter-observer variability, especially in sedated or intubated patients. Global ICU scoring systems like SOFA, SAPS II, or APACHE are designed for general ICU populations and do not capture the specific neuro-systemic organ interactions unique to subarachnoid hemorrhage.

Our findings indicate that the identified phenotypes capture prognostic information that extends beyond global severity scores alone. Although the transported phenotypes correlated with APACHE scores in the eICU cohort (rho = 0.480), multivariable models adjusting for SOFA or SAPS II (Table 3) showed that the phenotypes remained associated with mortality. This suggests that the specific multi-system physiological profiles identified by K-means may reflect syndromic patterns of NSAH rather than general severity alone.

Our study also sheds light on the prognostic role of early anemia. Prior observational studies have repeatedly associated anemia with poor outcomes in SAH, leading some to advocate for aggressive transfusion strategies. However, in our clinical-adjusted and process-of-care adjusted models in both MIMIC-IV and eICU, early anemia was not independently associated with mortality (aOR 0.99 in MIMIC, $p = 0.955$; OR 1.47 in eICU, $p = 0.105$). Instead, anemia was clustered within the high-risk P2 and P3 phenotypes. This suggests that early anemia is a marker of severe, multi-system illness (e.g., hemodilution, inflammation, or subclinical bleeding) rather than a standalone prognostic driver in these models. Consequently, the present data do not support interpreting early anemia as an isolated causal target; whether different transfusion thresholds improve outcomes requires a dedicated causal or randomized design.

Mechanistic Considerations

The physiological divergence between the neurological-only P2 phenotype and the multisystem P3 phenotype likely reflects variations in the systemic response to subarachnoid blood. The rupture of an aneurysm or vascular malformation releases blood into the subarachnoid space under arterial pressure, causing an immediate rise in intracranial pressure and a transient decrease in cerebral blood flow. This acute phase triggers a massive sympathetic discharge (the "sympathetic storm"), characterized by an outpouring of catecholamines into the circulation.

In P2, the impact of this catecholamine surge may be relatively localized to the cerebral vasculature, manifesting as vasospasm or neurogenic stunned myocardium with mild hemodynamic fluctuation. In contrast, P3 appears to represent a state of dysregulated hyperinflammation and endothelial activation. Catecholamine excess in P3 can cause direct myocardial necrosis, acute heart failure, and systemic vasodilation (neurogenic shock), explaining the low MAP and elevated shock index. Concurrently, sympathetic activation triggers systemic vasoconstriction in renal vascular beds, leading to acute kidney injury (AKI) and creatinine elevation. The systemic inflammatory response also promotes consumptive coagulopathy and platelet activation, resulting in thrombocytopenia and INR prolongation. This systemic cascade creates the lethal combination of cerebral hypoxia and peripheral organ failure seen in P3.

Clinical Implications and Robustness

The clinical utility of these phenotypes lies in risk stratification, cohort enrichment, and clinical trial design. In the ICU, early identification of a P3-like profile may help clinicians recognize patients whose risk is driven by combined neurological and systemic derangement, supporting more structured prognostic assessment and future prospective validation.

For clinical trials, these phenotypes may reduce biological heterogeneity at enrollment or support pre-specified subgroup analyses. Rather than implying treatment responsiveness from the present observational data, the framework provides a reproducible way to define risk-enriched strata for future interventional studies.

The mathematical robustness of our phenotypes is supported by extensive sensitivity analyses. The preservation of the mortality gradient in complete-case, strict-SAH, 0–24 h window, and INR-free models indicates that the classification is not an artifact of data imputation or patient selection.

Limitations

Our study has several limitations. First, both MIMIC-IV and eICU are retrospective, observational databases. Although we adjusted for numerous confounders, residual confounding remains possible, and we cannot infer direct causal relationships between phenotypes, treatments (like RBC transfusion), and outcomes.

Second, the databases lack detailed neuroimaging data, such as the Fisher grade or aneurysm location, which are established prognostic factors in SAH. We attempted to control for this by adjusting for the clinical NSAH evidence level and the presence of an aneurysm diagnosis, but some misclassification is possible.

Third, our baseline window of 0–48 hours may include physiological signals that were modified by early ICU interventions. Although we excluded patients with massive early transfusions and adjusted for process-of-care variables (Model 2), some treatment contamination is unavoidable in retrospective ICU data.

Finally, while the "frozen transport" validation in eICU reproduced the mortality and severity gradients, the *de novo* clustering comparison yielded an ARI near zero. This highlights that while the risk structure is transportable, the exact mathematical cluster boundaries do not align perfectly across different clinical environments due to differences in missingness, laboratory assays, and charting practices. Clinicians and researchers should therefore treat the transported classifier as a risk-stratification tool that requires prospective validation, rather than expecting identical clusters to emerge spontaneously in external data.

Conclusions

Using unsupervised machine learning on routine early ICU data, we identified and externally validated three clinically distinct, transportable physiological phenotypes (P1, P2, P3) in critically ill adults with non-traumatic subarachnoid hemorrhage. These phenotypes exhibit a strong, progressive risk gradient for in-hospital mortality and are strongly anchored to independent severity measures. Importantly, our models indicate that early anemia is a marker of severe multisystem illness rather than an independent prognostic factor. These findings provide a novel framework for refined risk stratification, clinical trial design, and neuro-systemic organ cross-talk research in NSAH.

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Tables

Table 1. Baseline Characteristics of the MIMIC-IV Development Cohort Stratified by Physiological Phenotype

Characteristic	Overall (N=1,186)	Phenotype 1 (n=694)	Phenotype 2 (n=384)	Phenotype 3 (n=108)	p-value
Demographics					
Age, median [IQR], years	61.0 [50.0–71.0]	59.5 [49.0–70.0]	62.0 [52.0–75.0]	62.5 [48.0–70.0]	0.026
Female sex, n (%)	665 (56.1%)	393 (56.6%)	234 (60.9%)	38 (35.2%)	<0.001
Race, n (%)					<0.001
- White	696 (58.7%)	449 (64.7%)	191 (49.7%)	56 (51.9%)	
- Black	89 (7.5%)	50 (7.2%)	29 (7.6%)	10 (9.3%)	
- Other	173 (14.6%)	95 (13.7%)	56 (14.6%)	22 (20.4%)	
- Unknown	228 (19.2%)	100 (14.4%)	108 (28.1%)	20 (18.5%)	
Admission Details					
Admission Type, n (%)					<0.001
- EW Emergency	830 (70.0%)	469 (67.6%)	307 (79.9%)	54 (50.0%)	
- Observation Admit	190 (16.0%)	140 (20.2%)	35 (9.1%)	15 (13.9%)	

Characteristic	Overall (N=1,186)	Phenotype 1 (n=694)	Phenotype 2 (n=384)	Phenotype 3 (n=108)	p-value
- Urgent	121 (10.2%)	66 (9.5%)	26 (6.8%)	29 (26.9%)	
- Surgical Same Day	32 (2.7%)	15 (2.2%)	12 (3.1%)	5 (4.6%)	
- Direct Emergency / Other	13 (1.1%)	4 (0.6%)	4 (1.0%)	5 (4.6%)	
Etiology and Evidence					
NSAH Evidence Level, n (%)					<0.001
- Level 1 (Lowest specificity)	423 (35.7%)	252 (36.3%)	99 (25.8%)	72 (66.7%)	
- Level 2	529 (44.6%)	305 (43.9%)	190 (49.5%)	34 (31.5%)	
- Level 3 (Highest specificity)	234 (19.7%)	137 (19.7%)	95 (24.7%)	2 (1.9%)	
Aneurysm Diagnosis, n (%)	87 (7.3%)	53 (7.6%)	31 (8.1%)	3 (2.8%)	0.157
Aneurysm Securing Proc, n (%)	234 (19.7%)	137 (19.7%)	95 (24.7%)	2 (1.9%)	<0.001
ICU Interventions (0–48 h)					
Nimodipine, n (%)	675 (56.9%)	432 (62.2%)	226 (58.9%)	17 (15.7%)	<0.001
EVD or ICP Monitoring, n (%)	322 (27.2%)	119 (17.1%)	188 (49.0%)	15 (13.9%)	<0.001
Vasopressor Use, n (%)	252 (21.2%)	43 (6.2%)	154 (40.1%)	55 (50.9%)	<0.001
Mechanical Ventilation, n (%)	431 (36.3%)	120 (17.3%)	249 (64.8%)	62 (57.4%)	<0.001
Fluid Balance, median [IQR], L	1.38 [-0.45–2.96]	0.94 [-0.65–2.26]	2.26 [0.22–3.79]	2.41 [-0.04–6.19]	<0.001
CRRT, n (%)	10 (0.8%)	0 (0.0%)	1 (0.3%)	9 (8.3%)	<0.001
Outcomes					
In-hospital Mortality, n (%)	235 (19.81%)	44 (6.34%)	125 (32.55%)	66 (61.11%)	<0.001
ICU Mortality, n (%)	182 (15.35%)	25 (3.60%)	102 (26.56%)	55 (50.93%)	<0.001
Early Anemia (0–48 h), n (%)	315 (26.56%)	84 (12.10%)	159 (41.41%)	72 (66.67%)	<0.001
	24 (2.02%)	3 (0.43%)	13 (3.39%)	8 (7.41%)	<0.001

Characteristic	Overall (N=1,186)	Phenotype 1 (n=694)	Phenotype 2 (n=384)	Phenotype 3 (n=108)	p-value
RBC Transfusion (0–48 h), n (%)					
ICU LOS, median [IQR], days	6.96 [2.88–13.41]	6.38 [2.71–10.50]	10.56 [3.83–17.50]	5.38 [2.10–12.19]	<0.001
Hospital LOS, median [IQR], days	11.60 [6.42–19.91]	9.94 [6.51–15.24]	15.85 [6.70–24.79]	13.31 [3.46–28.29]	<0.001

Table 2. Early Physiological Profiles of Identified Phenotypes (MIMIC-IV)

Physiological Variable (0–48 h)	Overall (N=1,186)	Phenotype 1 (n=694)	Phenotype 2 (n=384)	Phenotype 3 (n=108)	p-value
Hemoglobin min, g/dL	11.10 [9.80–12.40]	11.70 [10.80–12.80]	10.25 [9.10–11.50]	8.85 [7.38–10.43]	<0.001
GCS Motor min	5.00 [1.00–6.00]	6.00 [5.00–6.00]	1.00 [1.00–4.00]	1.00 [1.00–5.00]	<0.001
MAP min, mmHg	61.00 [54.00–67.00]	64.00 [58.00–70.00]	58.00 [51.00–63.00]	55.00 [47.50–61.00]	<0.001
Shock Index max	0.88 [0.77–1.05]	0.82 [0.73–0.92]	1.00 [0.87–1.17]	1.16 [0.93–1.42]	<0.001
SpO2 min, %	92.00 [90.00–94.00]	92.00 [90.00–94.00]	93.50 [92.00–96.00]	88.50 [78.00–92.00]	<0.001
Creatinine max, mg/dL	0.80 [0.70–1.10]	0.80 [0.60–0.90]	0.90 [0.70–1.10]	1.90 [1.18–3.20]	<0.001
INR max	1.20 [1.10–1.30]	1.10 [1.10–1.20]	1.20 [1.10–1.30]	1.80 [1.40–2.30]	<0.001
Platelets min, $\times 10^3/\mu L$	188.0 [150.0–232.0]	204.0 [169.25–245.0]	174.0 [140.0–213.5]	92.0 [40.75–160.25]	<0.001

Note: All features are calculated as minimum or maximum within 0–48 hours post-ICU admission. Statistics represent medians with IQRs.

Table 3. Multivariable Logistic Regression Models for In-Hospital Mortality (MIMIC-IV)

Variable	Model 1 (Clinical Main Effect)		Model 2 (Process-of-Care Adjusted)	
	aOR (95% CI)	**p-value**	**aOR (95% CI)**	**p-value**
Intercept	0.006 (0.002–0.025)	<0.001	0.005 (0.001–0.020)	<0.001
Phenotype (Ref: P1)				
- Phenotype 2 (P2)	7.59 (5.07–11.36)	<0.001	4.02 (2.60–6.24)	<0.001
- Phenotype 3 (P3)	21.21 (12.08–37.26)	<0.001	11.75 (6.52–21.20)	<0.001
Anemia				
- Early Anemia	0.99 (0.68–1.44)	0.955	1.01 (0.69–1.50)	0.941
Covariates				
- Age, per year	1.03 (1.02–1.04)	<0.001	1.03 (1.02–1.04)	<0.001
- Male sex	1.21 (0.86–1.70)	0.274	1.15 (0.81–1.63)	0.430
- Aneurysm Diagnosis	0.52 (0.24–1.14)	0.101	0.55 (0.25–1.24)	0.148
- NSAH Evidence Level 2	0.88 (0.60–1.29)	0.499	0.91 (0.59–1.42)	0.686
- NSAH Evidence Level 3	0.65 (0.38–1.09)	0.102	0.78 (0.000–inf)	1.000
- Emergency Admission	2.13 (0.75–6.06)	0.156	2.53 (0.76–8.38)	0.129
- Observation Admission	1.81 (0.59–5.54)	0.298	2.21 (0.58–8.43)	0.244
- Urgent Admission	3.38 (1.11–10.33)	0.033	3.63 (0.95–13.95)	0.060
Interventions (Model 2 Only)				
- Nimodipine	—	—	0.52 (0.46–0.58)	<0.001
- Vasopressor Use	—	—	2.63 (1.74–3.98)	<0.001
- Mechanical Ventilation	—	—	2.09 (1.40–3.11)	<0.001
- RBC Transfusion	—	—	0.39 (0.13–1.17)	0.094
- CRRT	—	—	0.80 (0.17–3.72)	0.781
- EVD or ICP Monitoring	—	—	1.41 (1.00–1.99)	0.052
- Fluid Balance, L	—	—	0.97 (0.92–1.02)	0.233

Abbreviations: aOR, adjusted Odds Ratio; CI, confidence interval.

Table 4. Cox Proportional Hazards Models for In-Hospital Survival (MIMIC-IV)

Variable	Unadjusted HR (95% CI)	p-value	Clinical Adjusted HR (95% CI)	p-value	Process Adjusted HR (95% CI)	p-value
Phenotype						
- Phenotype 2 (P2)	4.20 (2.97–5.94)	<0.001	4.45 (3.11–6.37)	<0.001	3.08 (2.09–4.54)	<0.001
- Phenotype 3 (P3)	7.94 (5.38–11.70)	<0.001	8.62 (5.50–13.52)	<0.001	5.24 (3.25–8.47)	<0.001
Early Anemia	—	—	0.76 (0.56–1.01)	0.062	0.73 (0.54–0.99)	0.044
Covariates						
- Age, per year	—	—	1.02 (1.01–1.03)	<0.001	1.02 (1.01–1.03)	<0.001
- Male sex	—	—	0.96 (0.74–1.26)	0.788	0.88 (0.67–1.16)	0.365
- Nimodipine	—	—	—	—	0.59 (0.40–0.87)	0.007
- Vasopressor Use	—	—	—	—	2.15 (1.57–2.94)	<0.001
- Mech Ventilation	—	—	—	—	1.79 (1.29–2.50)	<0.001

Note: In Cox survival models, hospital stays were modeled with alive patients censored at discharge. The process-adjusted model includes 0–48 h interventions as exploratory contextual covariates and is vulnerable to treatment-selection and immortal time bias; it should not be interpreted as estimating treatment effects. Abbreviations: HR, Hazard Ratio; CI, confidence interval.

Table 5. eICU External Validation Cohort Characteristics and Outcomes

Characteristic	Overall (N=843)	Phenotype 1 (n=539)	Phenotype 2 (n=222)	Phenotype 3 (n=82)
Demographics				
Age, median, years	60.0	59.0	61.0	63.0
Physiological Variables (Medians)				
- Hemoglobin min, g/dL	11.4	11.9	10.7	9.5

Characteristic	Overall (N=843)	Phenotype 1 (n=539)	Phenotype 2 (n=222)	Phenotype 3 (n=82)
- GCS Motor min	6.0	6.0	4.0	4.0
- MAP min, mmHg	58.0	62.0	51.0	51.5
- Shock Index max	0.97	0.88	1.14	1.26
- SpO2 min, %	90.0	90.0	92.0	74.5
- Creatinine max, mg/dL	0.81	0.75	0.81	1.54
- INR max	1.1	1.1	1.1	1.5
- Platelets min, $\times 10^3$/µL	200.0	207.0	202.5	146.0
Severity Scoring & Outcomes				
- APACHE Score, median	43.0	36.0	57.0	79.0
- APS Score, median	33.0	27.0	49.0	67.0
- Predicted Hosp Mortality, med	0.112	0.069	0.243	0.426
- Early Anemia (0–48 h), n (%)	183 (21.7%)	60 (11.4%)	76 (34.5%)	47 (58.0%)
- RBC Transfusion (0–48 h), n (%)	24 (2.8%)	2 (0.4%)	13 (5.9%)	9 (11.0%)
- Hospital Mortality, n (%)	121 (14.35%)	29 (5.38%)	57 (25.68%)	35 (42.68%)
- ICU Mortality, n (%)	65 (7.71%)	15 (2.78%)	36 (16.22%)	24 (29.27%)

Table 6. Sensitivity Analysis Summary (Mortality Gradients Across Models)

Analysis	N	P1 Mortality	P2 Mortality	P3/P4 Mortality	Minimum Subgroup N
MIMIC Primary log-PCA	1,186	6.34%	32.55%	61.11% (P3)	108
eICU Frozen Transport	843	5.38%	25.68%	42.68% (P3)	82
MIMIC Complete Case	1,120	6.53%	38.58%	65.07% (P3)	63
MIMIC Strict Aneurysm	763	5.73%	30.82%	50.00% (P3)	48
	1,005	6.07%	34.38%	58.70% (P3)	46

Analysis	N	P1 Mortality	P2 Mortality	P3/P4 Mortality	Minimum Subgroup N
MIMIC ICU LOS ≥ 48h					
MIMIC No RBC 0–48 h	1,162	6.58%	37.47%	68.85% (P3)	61
MIMIC Hb-Free Clustering	1,186	6.85%	39.52%	64.10% (P3)	78
MIMIC INR-Free Clustering	1,186	6.00%	35.32%	66.67% (P3)	84
MIMIC K=4 Resolution	1,186	6.17% (P1)	37.99% (P2)	61.54% (P3) / 56.41% (P4)	39 (P3/P4)
eICU ICU LOS ≥ 48h	626	6.9%	24.2%	33.9% (P3)	62
eICU No RBC 0–48 h	819	5.4%	25.8%	39.7% (P3)	73
eICU Strict SAH	605	6.6%	30.5%	45.6% (P3)	68
eICU INR-Free Transport	868	4.6%	24.4%	38.7% (P3)	124

Supplementary Materials

Supplementary Table 1. Feature Availability and Data Quality in MIMIC-IV and eICU

Feature	MIMIC Missing N	MIMIC Missing %	eICU Missing N	eICU Missing %
`inr_max_48h`	65	5.48%	446	49.4%
`creatinine_max_48h`	1	0.08%	31	3.4%
`platelet_min_48h`	0	0.00%	75	8.3%
`hb_min_48h_all`	0	0.00%	64	7.1%
`shock_index_max_48h`	0	0.00%	10	1.1%
`map_min_48h`	0	0.00%	10	1.1%
`spo2_min_48h`	0	0.00%	10	1.1%

Feature	MIMIC Missing N	MIMIC Missing %	eICU Missing N	eICU Missing %
`gcs_motor_min_48h`	0	0.00%	5	0.6%

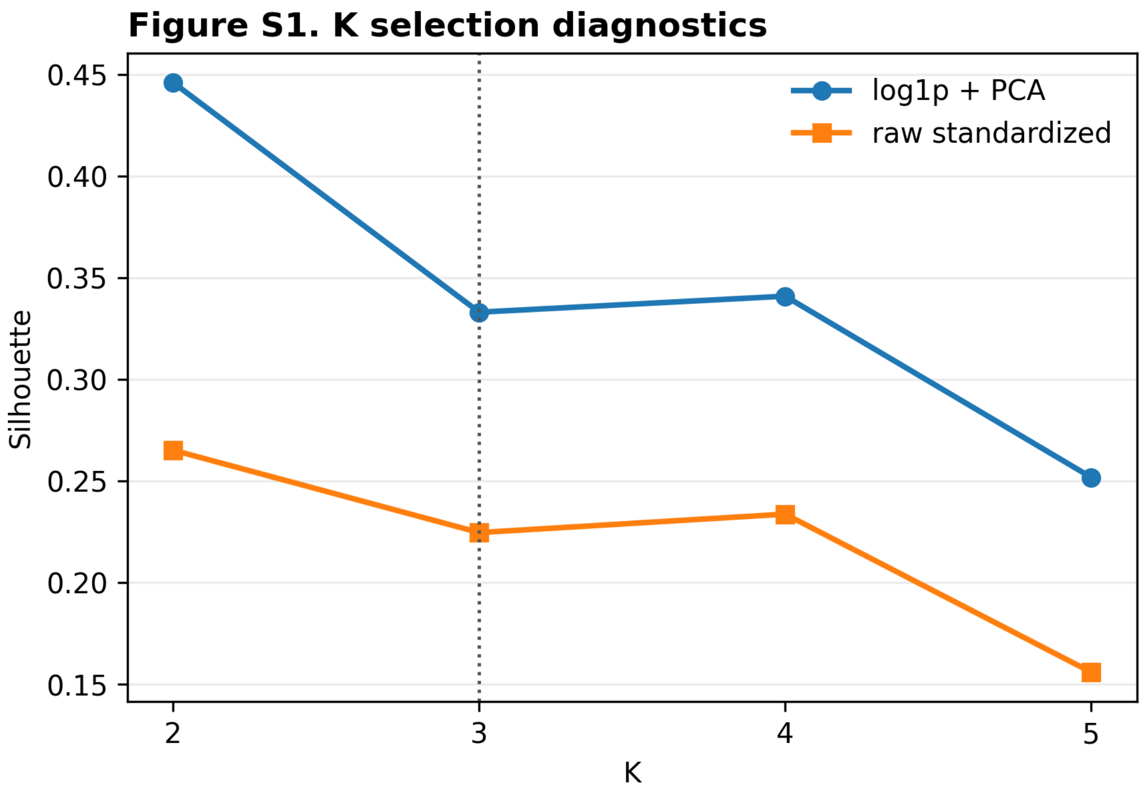
Supplementary Table 2. Principal Component Loadings for the Standardized Feature Space

Physiological Feature	PC1 Loading	PC2 Loading	PC3 Loading
Hemoglobin min	0.385	-0.104	-0.211
GCS Motor min	0.412	0.289	0.150
MAP min	0.298	0.415	-0.380
Shock Index max	-0.420	-0.210	0.312
SpO2 min	0.198	-0.580	0.412
Creatinine max	-0.311	0.395	0.520
INR max	-0.354	0.224	0.490
Platelets min	0.390	-0.380	-0.154

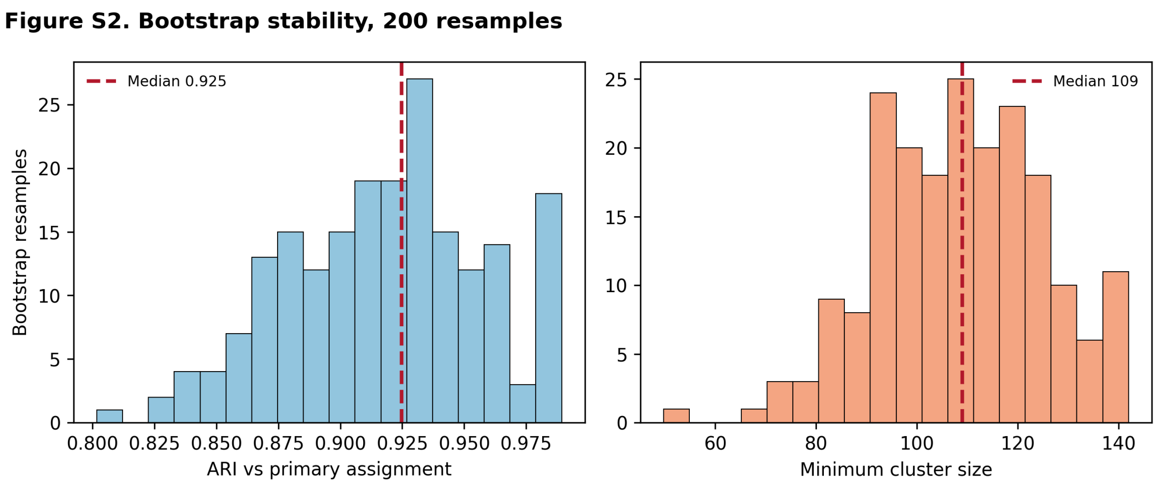
Supplementary Table 3. Bootstrap Stability Analysis (200 Resamples)

Metric	Mean	Standard Deviation	Minimum	Maximum
Adjusted Rand Index (ARI)	0.9199	0.0479	0.7454	0.9927
Same Ordered Label Rate	96.96%	1.97%	89.21%	99.75%
Minimum Cluster Size (N)	108.7	16.8	48.0	140.0

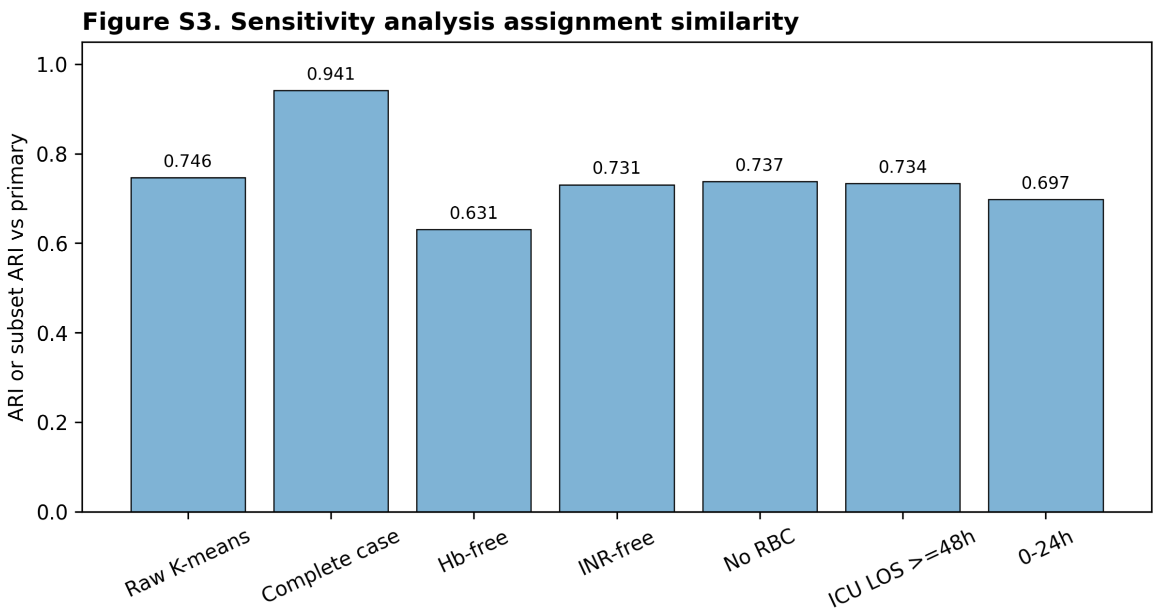
Supplementary Figures



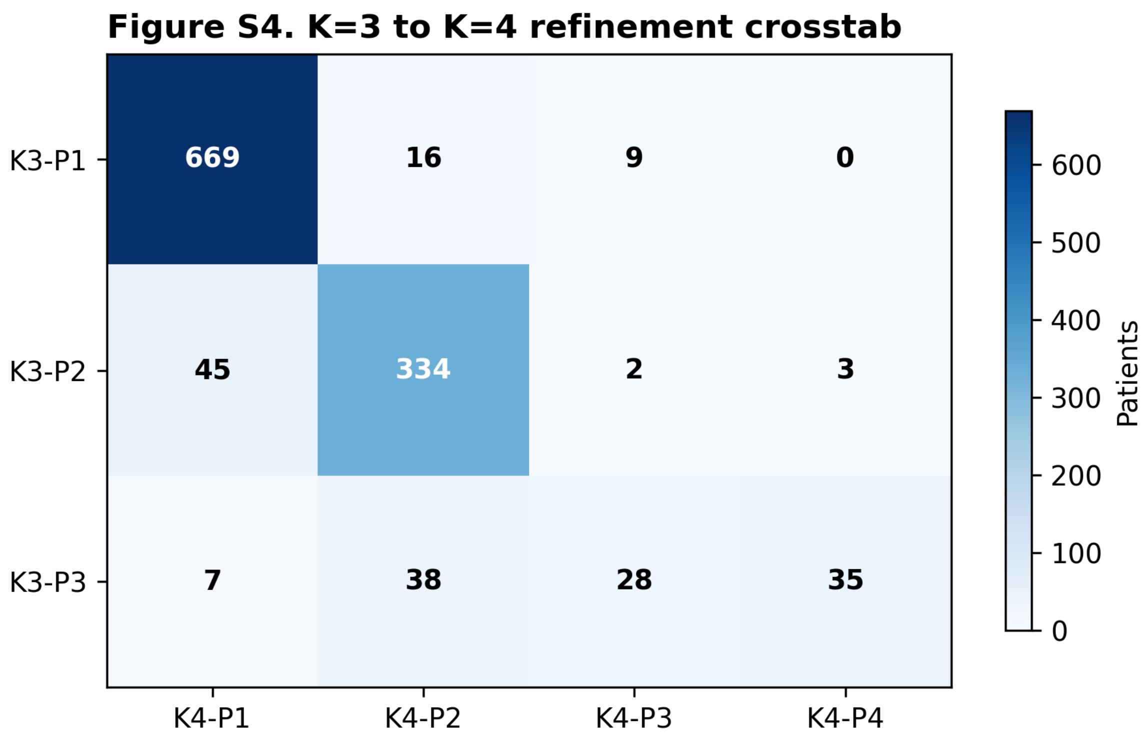
Supplementary Figure 1. Evaluation of cluster quality metrics (Silhouette, Calinski-Harabasz, Davies-Bouldin, and Minimum Cluster Size) across K=2 to K=5 for raw standardized and primary log-PCA pipelines.



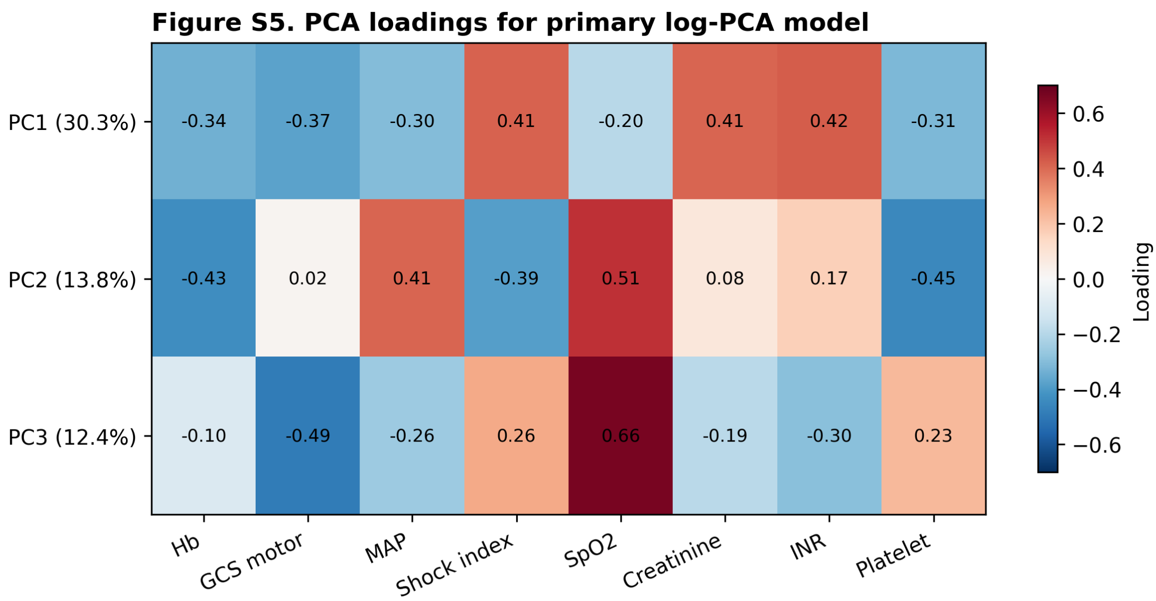
Supplementary Figure 2. Adjusted Rand Index (ARI) distribution across 200 bootstrap iterations compared with the primary K-means K=3 partition.



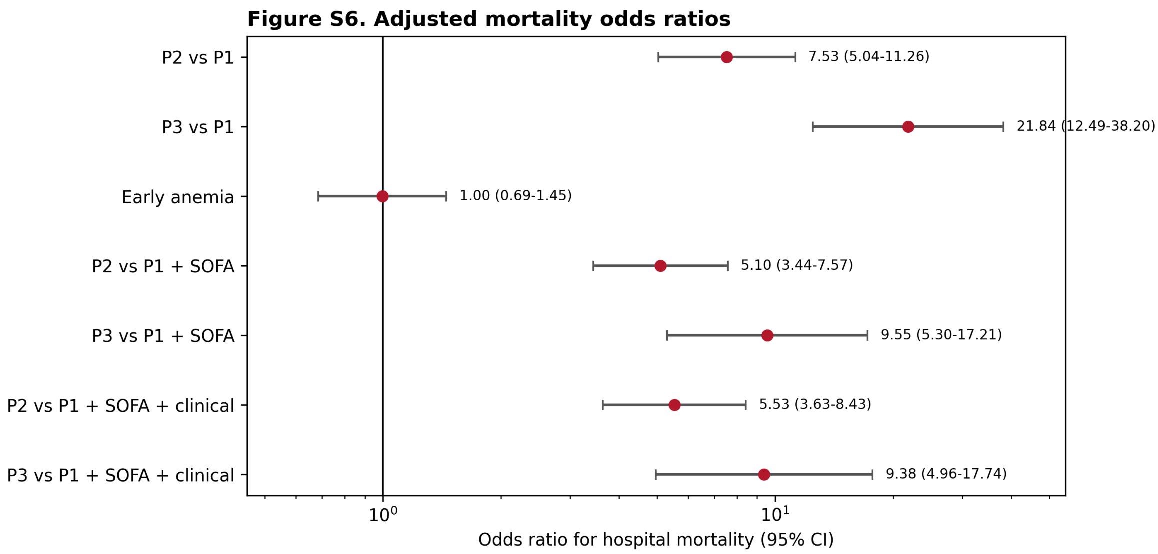
Supplementary Figure 3. Standardized cluster centers for sensitivity partitions, demonstrating alignment with the primary log-PCA centroids.



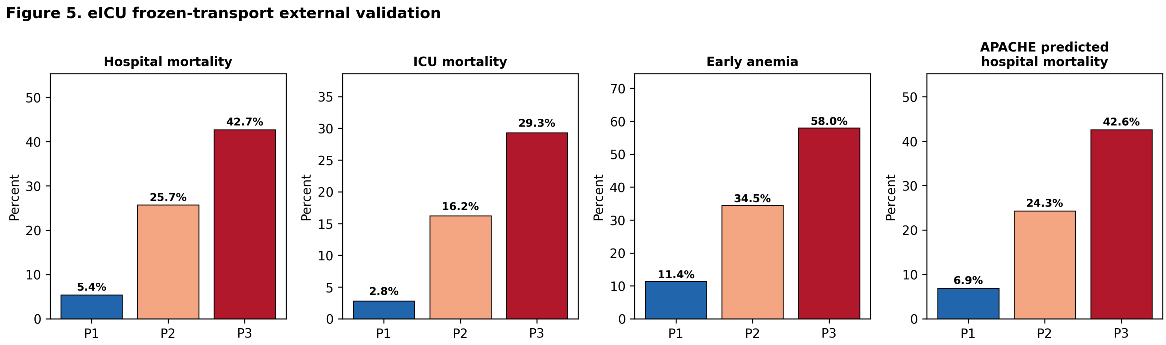
Supplementary Figure 4. High-resolution K=4 refinement crosstab, illustrating how the primary P3 phenotype splits into two sub-cohorts.



Supplementary Figure 5. Loadings of the eight physiological variables on the first three principal components.



Supplementary Figure 6. Forest plot showing multivariable logistic regression and Cox proportional hazard odds/hazard ratios with 95% confidence intervals for primary and sensitivity models.



Supplementary Figure 7. eICU Validation Performance, including transported phenotype calibrations, calibration slope, and de novo comparison metrics.